

Climate intervention techniques could reduce the nutritional value of crops

By Brendan Clark

A new study in [Environmental Research Letters](#) reports that cooling the planet by injecting sulfur dioxide into the stratosphere, a proposed climate intervention technique, could reduce the nutritional value of the world's crops.

Scientists at Rutgers University used global climate and crop models to estimate how stratospheric aerosol intervention (SAI), one type of solar geoengineering, would impact the protein level of the world's four major food crops: maize, rice, wheat, and soybeans. The SAI approach, inspired by volcanic eruptions, would involve releasing sulfur dioxide into the stratosphere. This gas would transform into sulfuric acid particles, forming a persistent cloud in the upper atmosphere that reflects a small part of the Sun's radiation, thereby cooling the Earth.

While these cereal crops are primarily sources of carbohydrates, they also provide a substantial share of dietary protein for large portions of the global population. Model simulations suggested that increased CO₂ concentrations tended to reduce the protein content of all four crops, while increased temperatures tended to increase the protein content of crops. Because SAI would stop temperatures from increasing, the CO₂ effect would not be countered by warming, and protein would decrease relative to a warmer world without SAI.

"SAI would not perfectly counteract the impacts of climate change; it would instead create a novel climate where the relationship between CO₂ and surface temperatures is decoupled. This would likely reduce the protein content of crops, and impact plant ecology in other ways we do not yet fully understand," said Brendan Clark, a former doctoral student in the Department of Environmental Sciences at the Rutgers School of Environmental and Biological Sciences (SEBS), and lead author on the study.

Models show that SAI would affect crop protein differently across regions, with the largest declines in nations that are already malnourished and protein deficient. The authors highlight that more field studies and model development are needed to make more informed decisions about SAI.

"Are we willing to live with all these potential impacts to have less global warming? That's the question we're trying to ask here," said Alan Robock, a Distinguished Professor of Climate Science in the Department of Environmental Sciences at SEBS, and a co-author of the study. "We're trying to quantify each of the potential risks and benefits so we can make informed decisions in the future."

Brendan Clark is now a Postdoctoral Associate at Cornell University. Other scientists on the study include Lili Xia, Assistant Research Professor at Rutgers, Sam Rabin of the NSF National Center for Atmospheric Research, Jose Guarin of NASA Goddard Institute for Space Studies, and Jonas Jägermeyr of Columbia University.